GENOMICS CLINICAL ACTION REPORT & PREVENTIVE SURVEILLANCE SUMMARY

Patient: Melinda Ehrle (Melinda_Ehrle) · **Date:** October 12, 2026

Primary Audience: Primary Care Physician (PCP) (Annual Physical, Preventive Lab Screening & Prescribing) · **Co-Routing:** Cardiology, Oncology, Hematology, Neurology & Clinical Pharmacy

Scope: 1653 Curated Variants (1129 Genes) + 41 Pharmacogenes

SECTION 1 · PRIORITY GENOMIC FINDINGS, CARRIER STATUS & CLINICAL IMPACT

GENE, SYSTEM & INHERITANCE	VARIANT, RSID & COORDINATE	ZYGOSITY & DEPTH	CLASSIFICATION & SCORES	CLINICAL IMPACT, PRIMARY CARE ACTION & ZYGOSITY CONTEXT
POLG Heart & Cardiovascular **Autosomal Dominant**	**p.Gly737Arg · c.2209G>C · rs121918054** chr15:89323460	**Heterozygous** DP: 31x	**Pathogenic/Likely pathogenic** CADD 26.3 · AVI 30.3 · Conf 0.95	**POLG · Mitochondrial DNA Replication & Valproate Contraindication:** Pathogenic missense variant (p.Gly737Arg) in the catalytic subunit of mitochondrial DNA polymerase gamma (OMIM #174763), essential for mitochondrial genome replication. **Clinical Impact:** Heterozygous carriers are typically asymptomatic at baseline, but carry a **strict FDA Boxed Warning contraindication against valproic acid / divalproex sodium** due to risk of fatal valproate-induced hepatic necrosis; verify baseline AST/ALT on annual metabolic panels. **Heterozygous Susceptibility Allele:** Single-allele variant in a dominant/semi-dominant pathway; manage via guideline-directed surveillance without unprovoked invasive procedures.
TAT Multisystem **Autosomal recessive**	**p.Arg57Ter · c.169C>T · rs118203914** chr16:71576247	**Heterozygous** DP: 52x	**Pathogenic** CADD 36.0 · AVI 38.8 · Conf 0.95	**TAT · Hepatic Tyrosine Catabolism (Tyrosinemia Type II Carrier):** Premature stop-gained variant (p.Arg57Ter) in hepatic cytosolic tyrosine aminotransferase, the rate-limiting enzyme converting L-tyrosine to p-hydroxyphenylpyruvate, associated with autosomal recessive **Tyrosinemia Type II (Richner-Hanhart syndrome, OMIM #276600)**. **Clinical Impact:** Heterozygous carriers retain ~50% hepatic enzymatic activity via the wild-type allele, remain clinically asymptomatic on an unrestricted diet, and do **not** develop oculocutaneous keratitis or palmoplantar hyperkeratosis (no dietary tyrosine/phenylalanine restriction indicated; optional one-time fasting plasma tyrosine < 100 µmol/L confirms normal clearance). **Heterozygous Recessive Carrier:** Single-allele carrier with an intact wild-type parental allele (~50–100% baseline functional reserve); does not cause primary recessive disease and requires no invasive diagnostic workup.
ATM Multisystem **Autosomal dominant / AR**	**c.1236-2del · rs1565381646** chr11:108250699	**Heterozygous** DP: 18x	**Pathogenic/Likely pathogenic** CADD — · AVI — · Conf 0.71	**ATM · Homologous Recombination DNA Repair & Oncologic Surveillance:** Pathogenic canonical splice-acceptor deletion (c.1236-2del) in the ATM serine/threonine DNA double-strand break repair kinase (OMIM #114480). **Clinical Impact:** Heterozygous female carriers have a moderately elevated lifetime breast cancer risk (~20–30%) and increased pancreatic cancer risk; NCCN guidelines recommend annual breast MRI (starting age 30–35) and annual mammography, family history review for pancreatic screening, and minimizing unnecessary diagnostic ionizing radiation. **Heterozygous Susceptibility Allele:** Single-allele variant in a dominant/semi-dominant pathway; manage via guideline-directed surveillance without unprovoked invasive procedures.
VDR Multisystem **Pathway Modifier**	**c.-1172A>G · rs4516035** chr12:47906043	**Heterozygous** DP: 21x	**Likely pathogenic** CADD 5.4 · AVI 2.1 · Conf 0.68	**VDR · Calcitriol Receptor Transcription & Bone Mineral Homeostasis:** 5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine cholecalciferol (Vitamin D3) supplementation. **Heterozygous Carrier / Susceptibility Finding:** Single-allele

SECTION 2 · PREVENTIVE LABORATORY & CLINICAL SURVEILLANCE SCHEDULE (PCP ANNUAL PHYSICAL & SPECIALIST ROUTING)

Derived strictly from **Melinda Ehrle**'s variant callset for integration into the **Primary Care Physician (PCP) annual preventive physical**. Each row links the triggering **Gene, Variant, and Zygosity** to a specific laboratory panel or clinical evaluation. (Page 2 of 3)

GENE, VARIANT & ZYGOSITY	PHYSIOLOGICAL PATHWAY	LABORATORY TEST / EVALUATION	ORDERING CLINICIAN & CO-CONSULT	FREQUENCY	CLINICAL RATIONALE & THRESHOLDS
POLG **p.Gly737Arg · c.2209G>C · rs121918054** chr15:89323460 · DP: 31x **Heterozygous (Autosomal Dominant)**	**POLG · Mitochondrial DNA Replication & Valproate Contraindication** ClinVar: Pathogenic/Likely pathogenic	**Comprehensive Metabolic Panel (CMP — Hepatic AST, ALT, Bilirubin & eGFR)** Venous Blood Draw + Permanent EHR Valproate Allergy Flag	**Primary Care Physician (PCP)** Co-Consult: Neurology / Clinical Pharmacy	Annual PCP Preventive Blood Panel & permanent EHR prescribing alert	Pathogenic missense variant (p.Gly737Arg) in the catalytic subunit of mitochondrial DNA polymerase gamma (OMIM #174763), essential for mitochondrial genome replication. **Clinical Impact:** Heterozygous carriers are typically asymptomatic at baseline, but carry a **strict FDA Boxed Warning contraindication against valproic acid / divalproex sodium** due to risk of fatal valproate-induced hepatic necrosis; verify baseline AST/ALT on annual metabolic panels.
TAT **p.Arg57Ter · c.169C>T · rs118203914** chr16:71576247 · DP: 52x **Heterozygous (Autosomal recessive)**	**TAT · Hepatic Tyrosine Catabolism (Tyrosinemia Type II Carrier)** ClinVar: Pathogenic	**Fasting Plasma Tyrosine / Amino Acid Profile & Hepatic AST/ALT** Optional Baseline Venous Blood Draw	**Primary Care Physician (PCP)** Co-Consult: Medical Genetics (Reproductive Carrier Context)	Optional one-time baseline at routine PCP physical	Premature stop-gained variant (p.Arg57Ter) in hepatic cytosolic tyrosine aminotransferase, the rate-limiting enzyme converting L-tyrosine to p-hydroxyphenylpyruvate, associated with autosomal recessive **Tyrosinemia Type II (Richner-Hanhart syndrome, OMIM #276600)**. **Clinical Impact:** Heterozygous carriers retain ~50% hepatic enzymatic activity via the wild-type allele, remain clinically asymptomatic on an unrestricted diet, and do **not** develop oculocutaneous keratitis or palmoplantar hyperkeratosis (no dietary tyrosine/phenylalanine restriction indicated; optional one-time fasting plasma tyrosine < 100 µmol/L confirms normal clearance).
ATM **c.1236-2del · rs1565381646** chr11:108250699 · DP: 18x **Heterozygous (Autosomal dominant / AR)**	**ATM · Homologous Recombination DNA Repair & Oncologic Surveillance** ClinVar: Pathogenic/Likely pathogenic	**Annual Breast MRI & Tomosynthesis Mammogram (per NCCN) + CBC & CMP Blood Panel** High-Risk Breast Imaging & Venous Blood Draw	**Primary Care Physician (PCP)** & **High-Risk Oncology / Medical Genetics**	Annual Breast MRI & Mammogram (staggered q6m per NCCN)	Pathogenic canonical splice-acceptor deletion (c.1236-2del) in the ATM serine/threonine DNA double-strand break repair kinase (OMIM #114480). **Clinical Impact:** Heterozygous female carriers have a moderately elevated lifetime breast cancer risk (~20–30%) and increased pancreatic cancer risk; NCCN guidelines recommend annual breast MRI (starting age 30–35) and annual mammography, family history review for pancreatic screening, and minimizing unnecessary diagnostic ionizing radiation.
VDR **c.-1172A>G · rs4516035** chr12:47906043 · DP: 21x **Heterozygous (Pathway Modifier)**	**VDR · Calcitriol Receptor Transcription & Bone Mineral Homeostasis** ClinVar: Likely pathogenic	**Serum 25-Hydroxyvitamin D [25(OH)D], Total Calcium, Albumin & Alkaline Phosphatase** Venous Blood Draw	**Primary Care Physician (PCP)** Co-Consult: Endocrinology (if refractory deficiency)	Annual PCP Preventive Lab Panel	5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine cholecalciferol (Vitamin D3) supplementation.
MC1R	**MC1R ·**	**Annual Full-Body**	**Primary Care**	Annual	Established p.Asp294His red-hair-

SECTION 4 · ACTIONABLE PHARMACOGENOMIC PRESCRIBING GUARDRAILS (CPIC / DPWG MEDICATION & DIPLOTYPE REFERENCE)

PRESCRIBING ACTION TIER	AFFECTED MEDICATIONS BY CAUSAL PHARMACOGENE & DIPLOTYPE
🚫 AVOID / CONTRAINDICATED 1 Screened Drugs	**CYP2C19** (*2/*2) [Poor Metabolizer]: **Clopidogrel (Plavix)**
⬇️ REDUCE DOSE (DOSE DOWN) 1 Screened Drugs	**VKORC1** (*H6/*H6) [Normal / Indeterminate]: **Warfarin (Coumadin)**
⚠️ USE WITH CAUTION 96 Screened Drugs	**HDAC1** (*1/*1) [Normal Metabolizer]: **Corticosteroids For Systemic Use** · **IL13** (*1/*1) [Normal Metabolizer]: **Hepatitis Vaccines** **CASP3** (*1/*1) [Normal Metabolizer]: **Immunoglobulins** · **TNFRSF1B/PADI4/PTPN22/PGLYRP4/FCGR2A/FCGR2B/TLR5/CTNNA2/IL1B/IL1RN/CERS6/STAT4/CTLA4/TLR2/CD14/CDKA** (*1/*1) [Normal Metabolizer]: **Tumor Necrosis Factor Alpha (Tnf-Alpha) Inhibitors** · **AOX1** (*1/*1) [Normal Metabolizer]: **Xk469** · **YBX1** (*1/*1) [Normal Metabolizer]: **Abiraterone**
🔍 THERAPEUTIC MONITORING 1 Screened Drugs	**RGS5/RXRG/NR1I2/ABCG2** (*1/*1) [Normal Metabolizer]: **Irinotecan**